

ECS on Fargate or ECS on EC2?

Quartz Software · B2B SaaS · Level 2

24 employees · 2 engineers Current step: portfolio 91/100 · needs_revision

 This is a synthetic consulting case created for cloud architecture practice. No real customer data is represented.

01 / DISCOVERY

Discovery

Business brief

A software vendor already uses ECS and needs to choose how to run its containers as the service grows.

What the client initially shared

- The client needs a justified cloud architecture decision and a feasible transition plan.
- Interview the client to establish workload, service targets, budget and operational constraints.

Stakeholders

Alex Morgan
Business sponsor

Customer impact and business priorities

Priya Shah
CTO

Feasibility, reliability, and engineering capacity

Sam Chen
Lead engineer

How the current system behaves and who operates it

Jordan Ellis
CFO

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reversibility

Questions and answers

My question

Can you describe the current workload and expected growth: average and peak number of ECS tasks, traffic pattern, CPU and memory needs, availability or latency targets, and how much you expect demand to grow over the next 12–24 months

Alex Morgan

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

CPU is steady but a nightly worker needs a specialized host configuration; the API does not. (Source: Lead engineer)

My question

What availability and latency targets does the API need to meet, and are there any explicit service-level objectives or customer commitments?

Alex Morgan

I don't have a reliable answer to that yet. Could you narrow the question to a business constraint or a specific measurement? If we aren't measuring it, we can discuss what evidence you need.

My question

When weekday demand doubles, how quickly must additional container capacity be available to handle the increased load without degrading the customer experience?

Alex Morgan

2 engineers maintain the product. Confirm who owns deployment, patching, incidents and recovery before committing to an option. Two platform engineers; maintain 99.9% service availability; estimate a full month including idle capacity. (Source: Engineering manager)

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

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What special host setup does the nightly worker need, and how many hours does it run each night?

Alex Morgan

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

CPU is steady but a nightly worker needs a specialized host configuration; the API does not. (Source: Lead engineer)

My question

What exactly makes the nightly worker need a special host? Does it need a specific EC2 instance type, GPU, local storage, or special system access?

Alex Morgan

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

CPU is steady but a nightly worker needs a specialized host configuration; the API does not. (Source: Lead engineer)

My question

What is the current monthly cost of running these ECS workloads, and is there a budget limit for the future setup?

Alex Morgan

Two platform engineers; maintain 99.9% service availability; estimate a full month including idle capacity. These are planning constraints; compare actual estimates including operations, data transfer and resilience. No supplier quote has been approved. (Source: Budget owner)

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

My question

How much time do the two platform engineers currently spend each month managing servers, patching, scaling, and incidents?

Alex Morgan

[Skip to content](#) maintain 99.9% service availability; estimate a full month including idle capacity. These are planning constraints; compare actual estimates including operations, data transfer and resilience. No supplier quote has been approved.

(Source: Budget owner)

2 engineers maintain the product. Confirm who owns deployment, patching, incidents and recovery before committing to an option. Two platform engineers; maintain 99.9% service availability; estimate a full month including idle capacity. (Source: Engineering manager)

My question

Can the nightly worker run separately from the API, using a different ECS compute setup?

Alex Morgan

Twelve container tasks run continuously; demand doubles for two hours each weekday. (Source: Lead engineer)

CPU is steady but a nightly worker needs a specialized host configuration; the API does not. (Source: Lead engineer)

Evidence requests

No evidence requests recorded yet.

Facts, assumptions, and unknowns

Type	Note	Source / confidence	Validation
Fact	The client needs a justified cloud architecture decision and a feasible transition plan.	Client brief · high	—
Fact	Interview the client to establish workload, service targets, budget and operational constraints.	Client brief · high	—
Fact	Twelve ECS container tasks run continuously under normal load	Alex morgan · medium	—
Fact	Demand doubles for approximately two hours every weekday	Alex morgan · medium	—
Fact	API CPU usage is relatively stead	Alex morgan · medium	—
Fact	A nightly worker requires a specialized host configuration that the API workloads do not	Alex morgan · medium	—
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Type	Note	Source / confidence	Validation
Fact	Two platform engineers maintain the product	Lead engineer · medium	—
Fact	Twelve ECS tasks run continuously under normal load	lead engineer · medium	—
Fact	Demand doubles for two hours each weekday	lead engineer · medium	—
Fact	Ownership of deployment, host patching, incident response, and recovery is not yet confirmed	lead engineer · high	Ask the engineering manager and platform engineers who handles each task and how much time it takes every month
Fact	The nightly worker needs a special host setup, but the API does not	lead engineer · high	—
Unknown	We do not know exactly how much time the engineers spend on patching, scaling, and incidents.	lead engineer · medium	—
Unknown	We do not know who handles deployments, patching, incidents, and recovery.	engineering manager · medium	—

02 / PROBLEM FRAMING

Problem framing

Stated Problem	A software vendor already uses ECS and needs to choose how to run its containers as the service grows.
Actual Problem	The client needs to choose the right ECS compute model as the service grows. The solution must balance cost, scalability, availability, and operational effort.
Business Impact	A poor choice could increase cloud costs, create more work for a small engineering team, or make it harder to maintain 99.9% availability as demand grows.
Symptoms	Twelve container tasks run continuously, and demand doubles for two hours each weekday. CPU usage is mostly steady. The nightly worker needs a special host setup, while the API does not.
Constraints	Only two platform engineers maintain the service. The architecture must support 99.9% availability. Costs must be compared over a full month, including idle capacity, operations, data transfer, and resilience. The nightly worker also needs a special host configuration.
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Uncertainties	We do not know the exact cost of each option, how much time the engineers spend managing infrastructure, or the exact host requirements of the nightly worker. These points could change the final recommendation.
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03 / REQUIREMENTS

Business and technical requirements

ID	Requirement	Priority	Source
R01	Maintain 99.9% service availability.	must	lad engineer
R02	Compare the cost over a full month, including idle capacity, operations, data transfer, and resilience.	must	lead engineer

04 / ARCHITECTURE OPTIONS

Architecture options

OPTION 1	
ECS on EC2 for all workloads	
Summary	Run the API and nightly worker on ECS using EC2 instances. Use Auto Scaling to add capacity when demand increases.
Components	ECS runs the containers on EC2 instances. The API runs as an ECS service behind a load balancer. An Auto Scaling Group provides EC2 capacity and adds instances when needed. The nightly worker runs on EC2 hosts that support its special configuration.
Advantages	Lower compute cost can be possible for steady workloads. Full control over the EC2 hosts. Supports the special host requirements of the nightly worker.
Disadvantages	The team must manage EC2 capacity, patching and scaling. Idle EC2 capacity may still cost money.
Skip to content	More operational work for a team of only two engineers.

Cost	Compute cost may be lower for the steady 24/7 workload, but the company also pays for idle capacity and engineering time to manage the hosts.
Operations	High. The team manages ECS and the underlying EC2 instances, including capacity, patching and recovery.
Reliability	99.9% availability is possible, but enough EC2 capacity must be available across failures and traffic increases.
Security	The team is responsible for securing and patching the EC2 hosts in addition to securing the containers and AWS configuration.
Implementation	not hard because the company already uses ECS, but EC2 capacity and scaling must be configured and maintained correctly.
Assumptions	The two engineers can manage EC2 hosts and Auto Scaling. The workload is steady enough for EC2 capacity to remain well utilized.
Risks	Poor capacity planning could cause unused capacity or not enough capacity during peaks. Host management could create additional operational work for the small team.
Tradeoff Rationale	This option fits the steady 24/7 workload and gives full host control for the nightly worker. It may also be cost-efficient if the EC2 capacity stays well used. However, it creates more work for the two-engineer team because they must manage capacity, patching and host recovery. Keeping 99.9% availability during traffic peaks also requires careful capacity planning.

OPTION 2

ECS on Fargate for all workloads

Summary	Run both the API and nightly worker using ECS Fargate. AWS manages the underlying compute capacity.
Skip to content	ECS runs the containers on Fargate. The API runs behind a load balancer and scales based on demand.

Additional tasks can start during the two-hour weekday peak.

The nightly worker also runs as a Fargate task.

Advantages

No EC2 hosts to manage or patch.
Easier scaling when demand increases.
Lower operational work for the two-engineer team.

Disadvantages

The steady 24/7 workload may cost more than well-used EC2 capacity.
Fargate provides less control over the underlying hosts.
The nightly worker may not support its required special host setup.

Cost

Infrastructure management cost is lower, but compute cost may be higher for the twelve tasks that run continuously.

Operations

Low. AWS manages the underlying compute infrastructure.

Reliability

Easier capacity management can reduce operational risk. The ECS service can run tasks across multiple Availability Zones.

Security

The team does not need to patch or manage EC2 hosts. Container, IAM and network security are still the team's responsibility.

Implementation

Low to medium. The API should be easier to move, but the nightly worker must first be tested for Fargate compatibility.

Assumptions

The nightly worker can run without direct control of the underlying host.

Risks

The nightly worker may not work on Fargate because it needs a special host configuration.
Fargate may also cost more for the steady workload

Tradeoff Rationale

This option reduces infrastructure management and makes scaling easier when demand doubles. This is a good fit for a small team of two engineers and can help support 99.9% availability. However, the nightly worker needs a special host setup, which may not be supported by Fargate. Fargate may also cost more for the twelve tasks that run continuously.

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Trade-off matrix

Dimension	ECS on EC2 for all workloads	ECS on Fargate for all workloads
Cost	4/5	3/5
Reliability	4/5	4/5
Performance	4/5	3/5
Scalability	3/5	3/5
Security	3/5	5/5
Operational Burden	2/5	5/5
Implementation Speed	3/5	3/5
Team Fit	2/5	5/5
Flexibility	5/5	3/5
Reversibility	4/5	4/5

Contextual ratings by the learner · 1–5, higher is more favorable · not measured benchmarks.

Requirement traceability

Requirement	Decision	Status	Rationale
R01	Run the ECS service across multiple Availability Zones with enough EC2 capacity and Auto Scaling.	Satisfied	Multiple Availability Zones reduce the impact of a single instance or zone failure. Auto Scaling helps maintain enough capacity when demand increases or an instance fails.
R02	Use a full-month cost model before final approval, including EC2 utilization, idle capacity, scaling, operations, data transfer and resilience costs.	At risk	EC2 may be cost-efficient for the steady 24/7 workload, but the final decision must be validated with real monthly estimates. The comparison must include idle capacity and operational effort, not only compute prices.

Recommendation

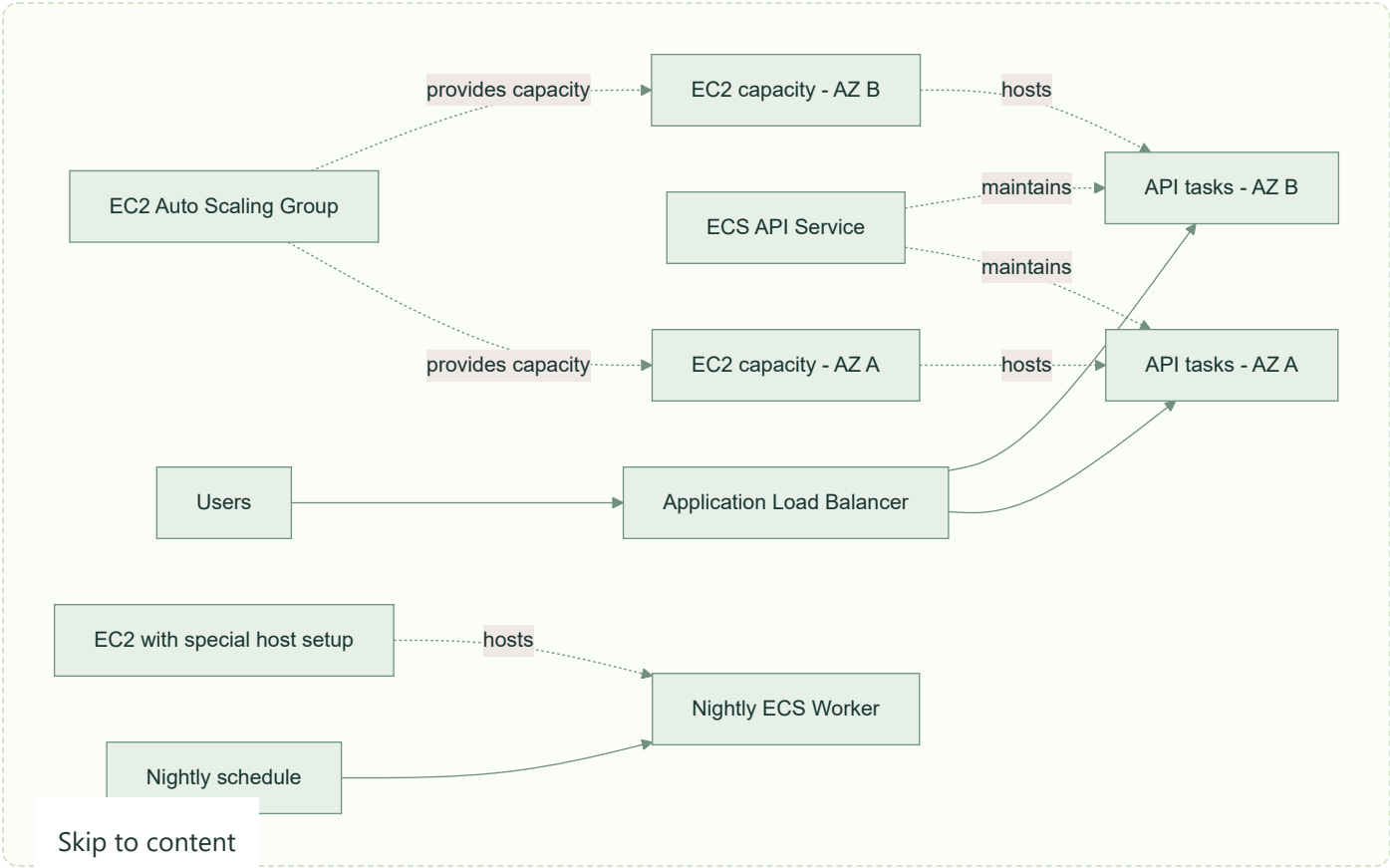
Selected Option ECS on EC2 for all workloads

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Opt 268c42ca-938e-4338-af85-d1eb58c2e269

Decision	Run the API and nightly worker on ECS with EC2. Use Auto Scaling to provide enough capacity during the weekday peak, and keep the special host configuration required by the nightly worker.
Why	The workload is mostly steady, with twelve tasks running all the time, so EC2 capacity can stay well used. EC2 also gives the team the host control required by the nightly worker. Auto Scaling can add capacity when demand doubles.
Alternatives	Fargate would reduce host management and make scaling easier, but the nightly worker needs a special host setup. We do not yet know if Fargate can support that requirement. Moving every workload to Fargate would therefore create unnecessary technical risk.
Risks	The two-engineer team will still need to manage EC2 capacity, patching and host recovery. Poor capacity planning could also create idle cost or reduce available capacity during traffic peaks.
Assumptions	We assume the twelve continuous tasks keep EC2 capacity reasonably utilized. We also assume the nightly worker really requires host-level control and that Auto Scaling can provide enough capacity for the weekday peak.

Architecture diagram



Why not something simpler?

The simpler option is to run everything on Fargate. It would reduce management and make scaling easier, but the nightly worker needs a special host setup. We therefore accept the extra EC2 management because we need host control and 99.9% availability.

What would change my mind?

01	Condition	The nightly worker can run on Fargate without needing a special EC2 host.
	Signal	Worker compatibility, performance, and reliability on Fargate.
	Threshold	The worker passes testing on Fargate with no host-related issues.
	Alternative	Move all ECS workloads to Fargate.

Validation and proof of concept

Required	Yes
Justification	Not recorded.
Hypothesis	ECS on EC2 can support the steady workload, handle the weekday traffic peak, and keep the nightly worker on its required host setup while meeting the 99.9% availability target.
Scope	Test the API across two Availability Zones, EC2 Auto Scaling during peak demand, and the nightly worker on the special EC2 host.
Measures	API availability, task health, EC2 utilization, scaling time, failed requests, worker success rate, and estimated monthly cost.
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Success	The API remains available during normal load and a 2x traffic test, new capacity is added when needed, and the nightly worker completes successfully on the special host.
Exit	Stop if the architecture cannot maintain the required availability, scaling is too slow, the worker fails, or the monthly cost is clearly higher than expected.
Go No Go	Go if the test meets availability, scaling, worker compatibility, and cost expectations. No-Go if one of these key requirements cannot be met.

10 / DECISION RECORD

ADR working draft

Title	ECS on EC2 for API and nightly worker
Context	The service runs twelve ECS tasks continuously, demand doubles for two hours each weekday, and the nightly worker needs a special host setup. The platform team has two engineers and must maintain 99.9% availability.
Decision	Run the workloads on ECS with EC2. Use an Auto Scaling Group across multiple Availability Zones for the API, and keep the nightly worker on EC2 with its required host configuration.
Alternatives	ECS on Fargate for all workloads was considered because it would reduce host management and make scaling easier. It was not selected because the nightly worker may require host control that Fargate cannot provide.
Consequences	The team keeps full host control and can support the special worker requirement. The trade-off is more operational work for EC2 capacity, patching, scaling and recovery.
Risks	The main risks are higher operational effort for the two-engineer team, poor capacity planning during traffic peaks, and higher cost if EC2 capacity is underused.

Accepted decision records

ADR-001 · ECS on EC2 for API and nightly worker

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The service runs twelve ECS tasks continuously, demand doubles for two hours each weekday, and the nightly worker needs a special host setup. The platform team has two

engineers and must maintain 99.9% availability.

Decision

Run the workloads on ECS with EC2. Use an Auto Scaling Group across multiple Availability Zones for the API, and keep the nightly worker on EC2 with its required host configuration.

Alternatives

ECS on Fargate for all workloads was considered because it would reduce host management and make scaling easier. It was not selected because the nightly worker may require host control that Fargate cannot provide.

Consequences

The team keeps full host control and can support the special worker requirement. The trade-off is more operational work for EC2 capacity, patching, scaling and recovery.

Risks

The main risks are higher operational effort for the two-engineer team, poor capacity planning during traffic peaks, and higher cost if EC2 capacity is underused.

Requirements

R01, R02

Reason

Not recorded.

Status

accepted

11 / COMMUNICATION

Communicating the decision

Engineer

We will run the API on ECS with EC2 across two Availability Zones. An Auto Scaling Group provides extra capacity when demand increases. The nightly worker stays on a special EC2 host because it needs host-level configuration. The main operational work is patching, scaling, monitoring and recovery.

Cto

This design keeps the special worker requirement while supporting 99.9% availability. It gives us more control than Fargate, but it also creates more operational work for a small team. We should review the decision if the worker can later run without special host access.

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Cfo	EC2 can be cost-efficient because most of the workload runs continuously. However, the real monthly cost must include idle capacity, extra resilience capacity and engineering time, not only compute prices. We should validate the full-month cost before final approval.
Ceo	The design is meant to keep the service available as the business grows and traffic increases. It also supports the special nightly workload without forcing a risky platform change. The trade-off is more infrastructure management for the engineering team.
Analogy	Think of EC2 like owning and managing your own building. You have more control and can customize it for special needs, but you are also responsible for maintenance. Fargate is more like renting a managed space: easier to operate, but with less control over the building.
Without Service Names	User traffic is distributed across application containers running in two separate locations for resilience. The system can add more server capacity when demand increases. A separate nightly workload runs on a specially configured server because it has different technical needs from the main API.

12 / REVIEW

Review and responses

Reviewer: Mock · local simulation

Decision-critical evidence · critical

A decision-critical operations constraint was not established. Discover it, or explicitly defer the decision with a measurable validation plan.

Response: accept

Rationale: I agree. We know the team is small, but we do not know the real operational workload. This could change the decision.

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Facts versus assumptions · warning

Your notes do not distinguish both assumptions and unknowns. Identify what is inferred, what remains unmeasured, and which uncertainty could change this decision.

Response: accept

Rationale: I agree. The traffic pattern and special host need are facts. EC2 utilization and operational effort are still assumptions or unknowns.

Accepted risk · warning

You marked R02 at risk. Explain the mitigation, owner, and threshold that would trigger an alternative.

Response: accept

Rationale: I agree. The budget owner should review a full-month cost estimate. If EC2 is more than 15% more expensive than Fargate, we should review the decision.

Validation · warning

No evidence request supports the recommendation yet. Defend why the available client statements are enough, or identify one measurement that could disprove your diagnosis.

Response: accept

Rationale: I agree. The recommendation still needs cost validation. A full-month EC2 cost estimate could prove the decision wrong.

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Trade-offs · info

You recommend “Run the API and nightly worker on ECS with EC2. Use Auto Scaling to provide enough capacity during the weekday peak, and keep the special host configuration required by the nightly worker.”. Defend it against your least complex alternative using a specific requirement and the team’s operating capacity. A more sophisticated service is not automatically a better fit.

Response: partial

Rationale: I partially agree. Fargate is simpler for the two-engineer team, but the nightly worker needs special host control. This is why EC2 is currently preferred.

Change of mind · info

Your first reconsideration threshold is “The worker passes testing on Fargate with no host-related issues.”. Explain who will observe this signal and when the decision will be reviewed.

Response: accept

Rationale: I agree. The platform engineers should test the worker on Fargate before production. If it works well, we should review the decision.

Communication · CEO / CFO · info

Your executive explanation includes service acronyms. Translate them into financial exposure, customer outcomes, and the trade-off the executive must approve.

Response: accept

Rationale: I agree. The CEO and CFO need business impact, cost and risk, not AWS service names.

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Final decision and lessons

Final Decision	I keep the ECS on EC2 recommendation because the nightly worker still needs special host control. After review, I would not approve production until we validate the real operational effort and full monthly cost. If the worker works well on Fargate or EC2 becomes much more expensive, I would review the decision.
Lessons	I learned that a recommendation is not enough by itself. I also need to explain how the decision will work in practice, what evidence will prove it, what success looks like, and what would make me change the decision.

EVALUATION

Evaluation

91/100

Needs revision

Transparent rubric · scenario rules

Critical misses

- A decision-critical operations constraint was not established. Discover it, or explicitly defer the decision with a measurable validation plan.

Dimension	Score	Feedback
Discovery quality	12/20	4/6 discovery topics established; 0 reasoned evidence request(s). Keep assumptions separate from facts and pursue constraints that could change the decision.
Problem framing	10/10	A strong framing separates the client’s symptoms from an actionable business problem and makes uncertainties explicit.
Architecture	15/15	2 alternatives compared; 100% of requirements traced. The rubric rewards documented fit, not particular cloud services.
Tr Skip to content	20/20	Trade-offs need explicit advantages, accepted costs, simpler alternatives, and connections to requirements. Numeric ratings do not determine this score.

Dimension	Score	Feedback
Cost and business reasoning	9/10	Tie costs to a stated budget, observed baseline, or financial impact. Ranges and explicitly labeled estimates are acceptable.
Security, reliability, operations	10/10	Document failure modes, security controls, operational ownership, and a practical validation or exit path.
Communication	10/10	Adapt the same decision to each audience. Executive explanations should describe business outcomes and risk in plain language.
Limits and change of mind	5/5	Revisit the decision when an observable threshold changes. A POC should have measurable success and stop criteria.

REASONING HISTORY

How the work changed

4 immutable snapshot(s) are saved.

First analysis

Final revision

Final revision

Final revision

Field	Before review	Final revision
Final Decision	Not recorded	I keep the ECS on EC2 recommendation because the nightly worker still needs special host control. After review, I would not approve production until we validate the real operational effort and full monthly cost. If the worker works well on Fargate or EC2 becomes much more expensive, I would review the decision.
Lessons	Not recorded	I learned that a recommendation is not enough by itself. I also need to explain how the decision will work in practice, what evidence will prove it, what success looks like, and what would make me change the decision.

PORTFOLIO

Publication status

Published in the portfolio.

This PDF reflects the saved case.